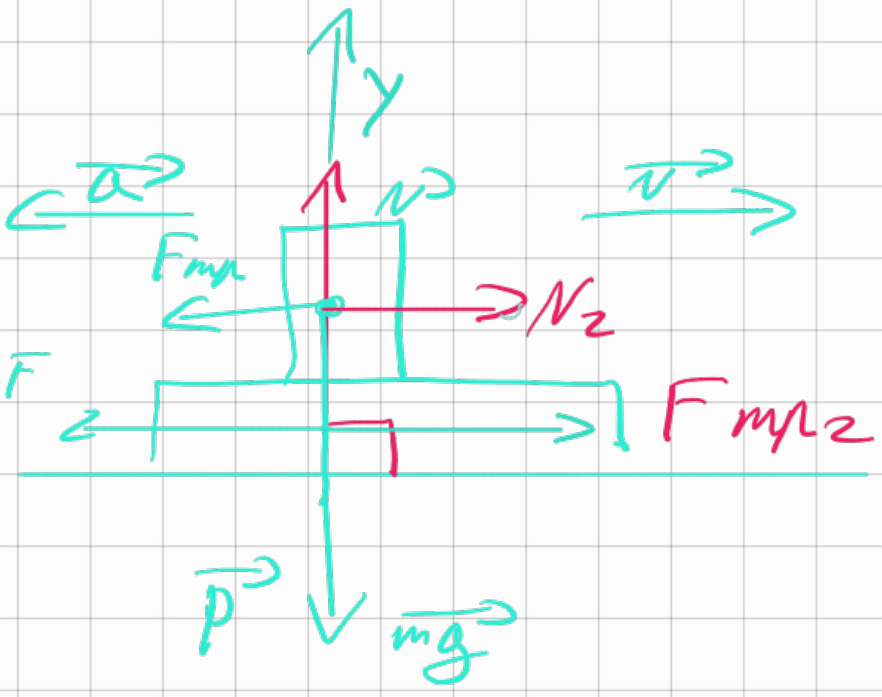
$m = 40$ кг
$L = 20$ м
$t = 10$ с
$F = 18$ Н

μ – ?
M – ?



$F_{тр1} = F_{тр2} = F_{тр}$

1 тело

$\vec{N} + \vec{F}_{тр1} + \vec{mg} = \vec{ma}$

2 тело

$\vec{F} + \vec{P} + \vec{Mg} + \vec{N_2} + \vec{F}_{тр2} = \vec{Ma}$

1

X: $F_{тр} = ma$

y: $N - mg = 0$

$$N = mg$$

$$F_{тр} = \mu N = \mu mg$$

$$\mu mg = ma$$

$$a = \mu g$$

2

$$x: \; F - F_{тр} = Ma$$

$$y: \; N_2 - P - Mg = 0$$

$$F - \mu mg = Ma$$

$$L = \frac{V^2 - V_0^2}{2a}$$

$$L = V_0 t + \frac{at^2}{2}$$

$$L = \frac{V_0 + V}{2} t$$

$$\mu = \frac{a}{g}; \; \mu = \frac{0{,}4}{10} = 0{,}04$$

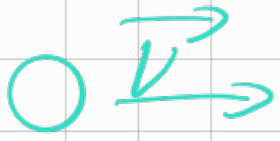
$$F = \mu \cdot N$$

m

v_0

$F = -kv^2$

$v = \frac{v_0}{2}$



$$F_c = ma$$

$$-kv^2 = ma$$

$$-kv^2 = m\frac{dv}{dt}$$

$$a = \frac{dv}{dt}$$

$$\frac{-k}{m}\int_0^t dt = \int_{v_0}^{v} \frac{dv}{v^2}$$

$$-\frac{k}{m}\int_0^t t^0\, dt = \int_{v_0}^{\frac{v_2}{2}} v^{-2}\, dv$$

$$-\frac{k}{m}t\Big|_0^{\tau} = -U^{-1}\Big|_{v_0}^{\frac{v_0}{2}}$$

$$-\frac{k}{m}t = -\left(\frac{2}{v_0} - \frac{1}{v_0}\right)$$

$$t = \frac{m}{v_0 k}$$

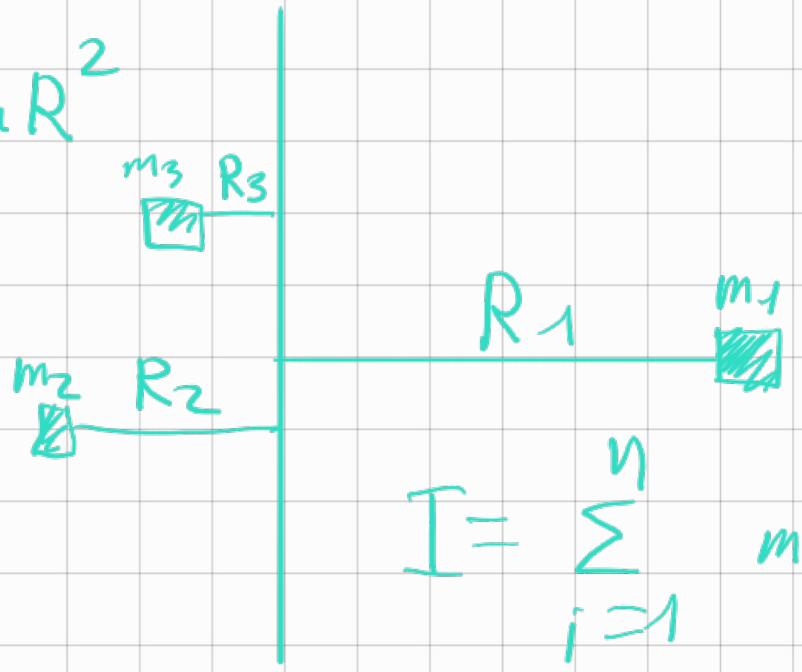
Динамика вращательного движения.

S	φ
V	ω
a	ε
x_0, v_0	φ_0, ω_0
$\vec{F} = m\vec{a}$	$\vec{M} = I \cdot \vec{\varepsilon}$
$\vec{p} = m\vec{V}$	
$\vec{F}\Delta t = \Delta\vec{p}$	

Моменты инерции (I)

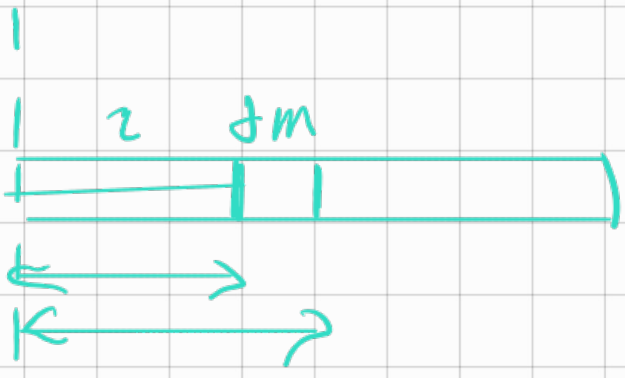
1. МТ

$$I = mR^2$$



$$I = \sum_{i=1}^{n} m_i R_i^2$$

$$I = \sum_{i=1}^{N} R^2 \Delta m = \int_V R^2 \, dm$$



l l

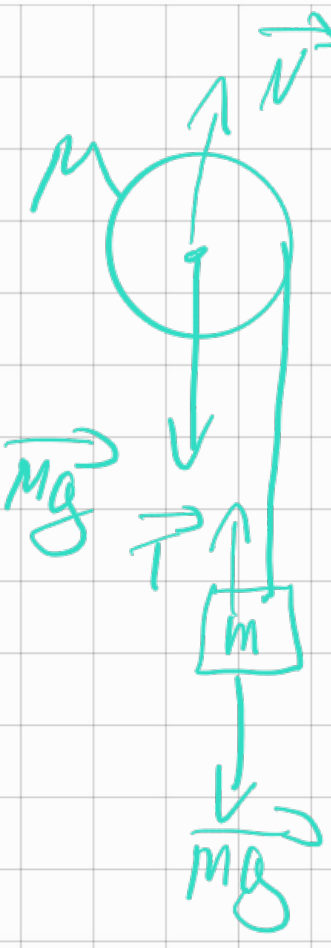
Дано:

$h = 15$ м

$M = 6$ кг

$m = 2$ кг

$t - ?$



1 тело

$\vec{T_1} + \vec{mg} = \vec{ma}$

$mg - T = ma$

2 тело

$\Sigma \vec{M} = I\vec{\varepsilon}$

$M_T = I \cdot \varepsilon$

$T \cdot R$